Fridge Photo Chef: Product Requirements Document

Owner: Tushara Hall | Version 1.0 | November 2, 2026 | Stack: Gemini API + YouTube Data API v3 + Netlify

1. Overview

Fridge Photo Chef is a web app that turns a photo of a fridge or pantry into cooking ideas. The user uploads a photo, the **Gemini API** identifies the ingredients, and the **YouTube Data API** finds recipe videos that use them. Videos play inside the app. The MVP is built and hosted for $0 on Netlify.

2. Problem & Goal

- **Problem:** People open the fridge, see random ingredients, and don't know what to cook, so food goes to waste or they order takeout.
- **Goal:** Go from photo to a playable recipe video in under 30 seconds.

3. Target Users

- Busy students and young professionals cooking on a budget
- Home cooks who want to use up leftovers and reduce food waste
- Beginner cooks who learn best from video

4. User Flow

1. User opens the app and taps **Upload or Take Photo**.
2. Gemini analyzes the photo and returns an ingredient list.
3. User reviews the list as editable chips (remove wrong items, add missing ones).
4. User optionally picks filters: meal type, max cook time, dietary preference.
5. User taps **Find Recipes**; the app searches YouTube with the ingredients.
6. App shows a grid of recipe videos (thumbnail, title, channel, length).
7. User taps a video; it plays in the app.

5. Functional Requirements (MVP)

ID	Requirement	API	Priority
F1	Upload a photo (JPG/PNG/HEIC, max 10 MB) or take one with the phone camera	None (front end)	Must
F2	Send the photo to Gemini and return ingredients as JSON (name + confidence)	Gemini image understanding + structured output	Must
F3	Show ingredients as editable chips; user can delete or add items	None	Must
F4	Build a search query from the top 3–5 ingredients plus "recipe"	None	Must

ID	Requirement	API	Priority
F5	Search YouTube for recipe videos (max 12 results, type = video, embeddable only)	YouTube search.list	Must
F6	Show video length and view count on each card	YouTube videos.list	Must
F7	Play the selected video inside the app	YouTube IFrame Player	Must
F8	Filters: meal type, under 15/30/60 min video length, vegetarian/vegan/high-protein	YouTube search params + Gemini	Should
F9	Gemini suggests 3 dish names from the ingredients; tapping one searches YouTube for it	Gemini text generation	Should
F10	"Ask about this video" (e.g. "What can I swap for butter?") using the video's URL	Gemini video understanding (YouTube URL)	Could
F11	Turn a chosen video into a written recipe card and shopping list for missing items	Gemini video understanding + structured output	Could
F12	Save favorite videos in the browser (no account)	None (localStorage)	Could

6. Non-Functional Requirements

Area	Requirement
Speed	Photo to results in under 30 seconds; show a loading state at each step
Security	API keys stored only in Netlify environment variables and used only inside Netlify Functions, never in front-end code
Privacy	Photos are not stored; show a note that the free Gemini tier may use inputs to improve Google products
Cost	$0: Gemini free tier, YouTube free quota, Netlify free plan
Quota	Cache YouTube results for repeat ingredient sets (24 hours) to stay under 10,000 units/day (about 100 searches)
Devices	Mobile-first and responsive; works on current Chrome, Safari, Edge and Firefox
Accessibility	Alt text, keyboard navigation, readable contrast (WCAG 2.1 AA)
Errors	Friendly messages for: no food detected, blurry photo, quota reached, no videos found

7. Technical Architecture

- **Front end:** HTML/CSS/JavaScript (or React) hosted on Netlify.
- **Function 1, /analyze-photo:** receives the image, calls Gemini (free-tier Flash model) with a JSON schema, returns ingredients.
- **Function 2, /find-videos:** receives ingredients and filters, calls YouTube search.list then videos.list, returns video cards.
- **Function 3, /ask-video (Could):** sends a YouTube URL and question to Gemini, returns the answer.
- **Environment variables:** GEMINI_API_KEY, YOUTUBE_API_KEY.

8. API Usage & Limits

API call	When	Cost / limit
Gemini image analysis	Once per photo	Free-tier rate limits (check before demos)
Gemini dish suggestions / video Q&A	Optional features	Free tier; YouTube URLs max 8 hours of video/day, public videos only
YouTube search.list	Once per search	100 units of 10,000/day
YouTube videos.list	Once per search (batch of IDs)	1 unit
YouTube IFrame Player	Each play	Free, no key

9. Success Metrics

- 80%+ of identified ingredients are correct (user deletes fewer than 1 in 5)
- 70%+ of sessions that upload a photo play at least one video
- Median photo-to-results time under 30 seconds
- Zero API keys exposed in front-end code

10. Out of Scope (v1)

- User accounts and cloud-saved favorites
- Grocery prices, store maps or delivery
- Nutrition or calorie counts
- AI-generated food images (paid-only Gemini feature)

11. Risks & Mitigations

Risk	Mitigation
Gemini misidentifies or misses items	Editable ingredient chips (F3); tips for clearer photos
YouTube quota runs out	Cache results; show "try again tomorrow" message
Videos don't match the ingredients well	Use top 3–5 ingredients; let Gemini suggest specific dish names (F9)
Free-tier limits change	Check the rate-limits page before demos; keep model name in one config value

12. Milestones

Milestone	Deliverable
M1: Setup	API keys, Netlify site, environment variables
M2: Photo to ingredients	F1–F3 working end to end
M3: Ingredients to videos	F4–F7 working; live Netlify URL
M4: Polish	F8–F9, error states, mobile testing
M5: Stretch	F10–F12

13. Documentation

- Gemini: image understanding • structured output • video understanding • rate limits • API key
- YouTube: search.list • videos.list • IFrame Player • quota costs
- Netlify Functions • Environment variables